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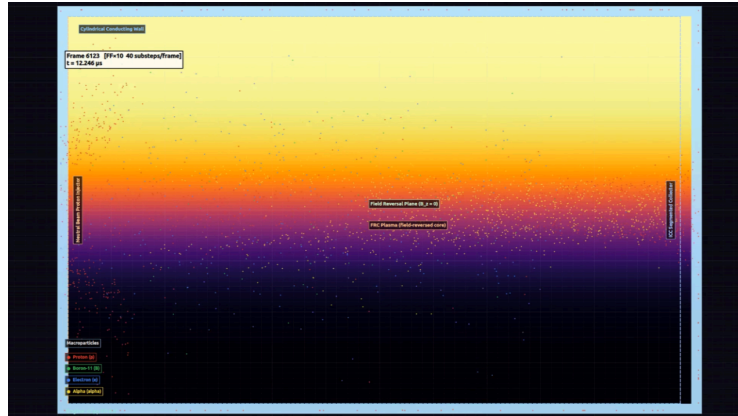
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Fusion and collection at full power

Simulating Proton–Boron Fusion in a TAE Field-Reversed Configuration

**Lars Warren Ericson** ✓

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June 5, 2026

Most fusion conversations start with deuterium–tritium. I have been exploring something harder and cleaner: **proton–boron-11 (p-11B)** — a reaction that produces **no neutrons**, only charged **helium-4 (alpha) particles** that can—in principle—be converted directly to electricity.

To make that tangible, we built **pb11_reactor_sim**: an open-source GUI that couples a **particle-in-cell (PIC) macroparticle view** to a **0D power-balance model** for three public reactor concepts. This article focuses on **TAE FRC** — a beam-driven **field-reversed configuration (FRC)** aligned with the direction **TAE Technologies, Inc** has described publicly, especially the **Norm** milestone (NBI-only FRC sustainment, 2025).

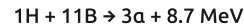
The simulator is **not** a CAD model of Norm. It is a **teaching slice** of the plasma core: NBI sustainment, p-11B fusion, and **inverse cyclotron converter (ICC)** recovery — exported as a narrated MP4 you can step through callout by callout.

Repository: github.com/catskillsresearch/pb11_reactor_sim

Disclaimer: *pb11_reactor_sim* was developed with AI assistance from public documentation. Models are for education and theoretical research — not operational specifications. See the repository **LICENSE** and **NOTICE** for full terms.

Why p-11B is a four-stage chain (not one line in a textbook)

It is tempting to write:

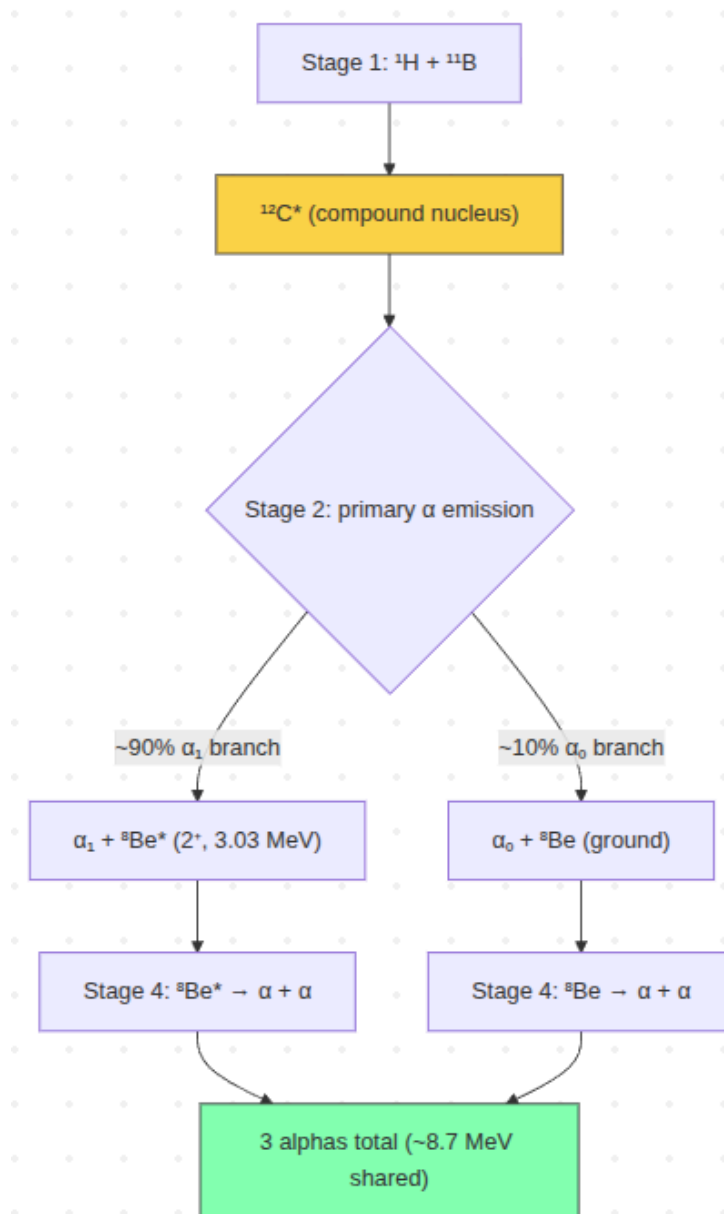


In reality, fusion proceeds through **short-lived intermediate nuclei**. That internal structure sets the **alpha energy spectrum** — which matters enormously for **direct energy conversion**, because a collector grid is an energy filter.

The four stages

1. $1\text{H} + 11\text{B} \rightarrow 12\text{C}^*$ — excited compound nucleus
2. $12\text{C}^* \rightarrow \alpha + 8\text{Be}^*$ — primary alpha emission
3. 8Be^* — unbound recoil nucleus
4. $8\text{Be}^* \rightarrow \alpha + \alpha$ — two secondary alphas

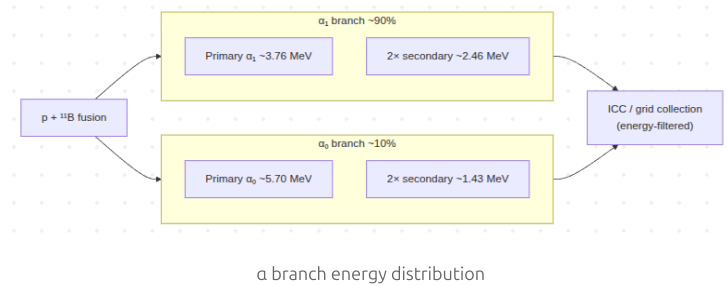
Net: three alphas sharing ~8.7 MeV, but **not** with equal energies.



The p-B11 fusion equation

Two branches compete at stage 2:

- **The α_1 branch (~90% probability):** $12\text{C}^* \rightarrow \alpha_1 + 8\text{Be}^*(2+, 3.03 \text{ MeV}) \rightarrow \alpha_1 + 2\alpha$ (yields a primary α_1 of ~3.76 MeV and two secondary alphas of ~2.46 MeV)
- **The α_0 branch (~10% probability):** $12\text{C}^* \rightarrow \alpha_0 + 8\text{Be}(\text{ground}) \rightarrow \alpha_0 + 2\alpha$ (yields a primary α_0 of ~5.70 MeV and two secondary alphas of ~1.43 MeV)



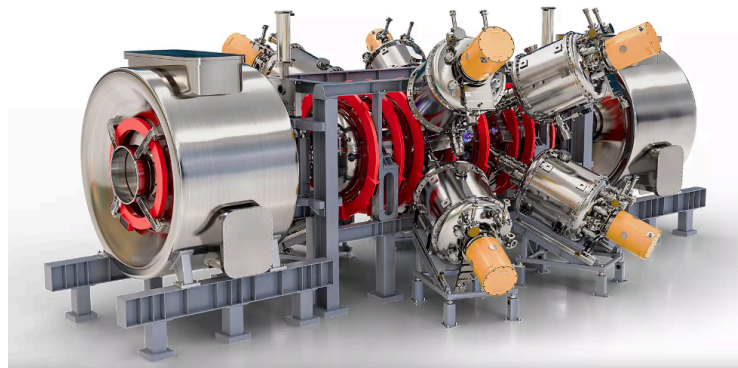
Intermediates live only **10-21–10-17 s** — far shorter than our 2 ns simulation timestep — so we do **not** transport 12C^* or 8Be as particles. The reaction is treated as instantaneous $\text{p} + 11\text{B} \rightarrow 3\alpha$, while **alpha birth energies** are sampled from the realistic multi-Gaussian spectrum (~2.9 MeV mean per alpha).

Why that matters for TAE: ICC and electrostatic collectors respond to **voltage sweeps** the way real hardware would — energetic primaries punch through higher retarding potentials; soft secondaries turn back earlier. Monoenergetic alphas would give an unphysical step-function collection curve.

The real machine (3D) vs. our video (2D)

What TAE is building (3D research accelerator context)

TAE's approach is a **compact linear FRC** — a rotating plasma structure whose internal magnetic field **reverses** relative to the applied field (often described as a “smoke ring” in cross-section).



TAE Norm

TAE Technologies Fusion Lab Tour

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Public milestones relevant here:

- **Norm (2025):** reported **neutral beam injection (NBI)**—only formation, heating, and sustainment of an FRC **without** older end **collision/formation** sections — a simplification TAE describes as a step toward commercial plants.
- **Earlier devices (e.g., Norman):** more elaborate end formation; still beam-driven FRC physics.
- **Roadmap names (Copernicus, Da Vinci):** aim at net energy and grid power with thermal wall recovery **plus** direct alpha conversion. **Those full plants are not simulated here.**

A full commercial design harvests energy two ways:

1. **Thermal** — radiation on walls → steam plant
2. **Direct** — fast alphas on open field lines → **ICC** at the ends

Our module keeps the ICC thread and omits steam turbines, divertors, vacuum plumbing, and powder-dropper hardware.

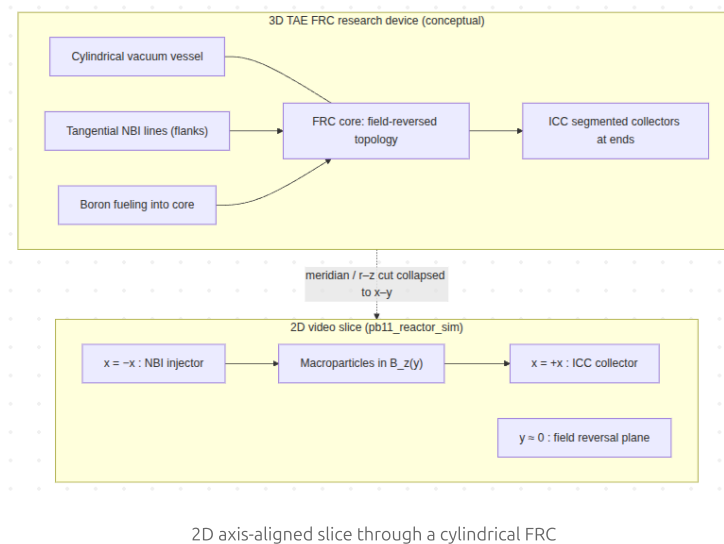
Two fuel paths (protons vs. boron)

p-11B needs **both** hydrogen and boron-11, but engineering separates injection:

- **Protons:** In TAE's FRC approach, they are introduced via tangential **NBI** from the machine's flanks. Protons are accelerated, neutralized, injected, and re-ionized inside the hot core, providing heat, rotation, and current drive. In the simulator, they are represented by **red** macroparticles entering from **−x**, where the **NBI Current** slider sets beam energy and rate.
- **Boron-11:** In TAE's FRC approach, Boron-11 is introduced directly into the **core** (as gas or powder), separate from NBI. In the simulator, these are represented by **green** macroparticles in the midplane, where the **gas fill** rate raises the core inventory.

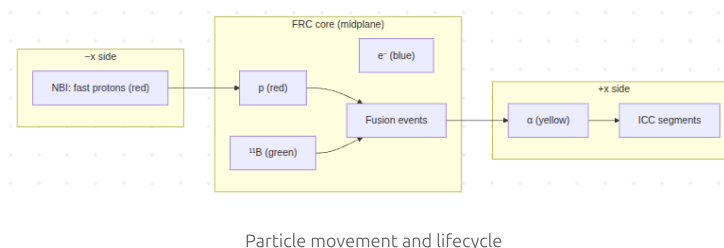
Mapping the 2D video view to the 3D machine

The simulation shows a **2D axis-aligned slice** through a cylindrical FRC — a **meridian cut** collapsed for visualization.



The geometric orientation of the 2D simulator coordinates maps directly to the physical 3D reactor design as follows:

- **Horizontal Axis (x):** Represents the **axial** direction along the length of the machine (~1.2 m window).
- **Vertical Axis (y):** Radial-like coordinate across the **field-reversal plane** (where $B_z = 0$ at the midplane).
- **Color map (B_z):** Represents the strength of the axial magnetic field; yellow (+B) above, purple (-B) below.
- **Left boundary (-x):** Region where the **NBI** enters (labeled *Neutral Beam Proton Injector*).
- **Right boundary (+x):** Represents the **ICC segmented collector** (containing 8 electrodes).
- **Top/bottom/left walls:** Represents the grounded **conducting chamber** outer shell.
- **Colored dots:** Represents **macroparticles** — where each dot represents a large swarm of real physical particles.
- **Cyan overlays:** Solid conductors (chamber walls and ICC segments) rather than moving plasma particles.



- **Red + green** = fuel; fusion where they overlap hot and dense
- **Yellow** = fusion product; born in the core, stream +x toward ICC
- **Blue** = electrons; their temperature **T_e** drives Bremsstrahlung losses

The **right-hand panel** is not spatial — it is a **time history** of the 0D power balance (temperatures, power terms, gain).

Control inputs and the optimum this model finds

Three inputs define the TAE FRC operating point:

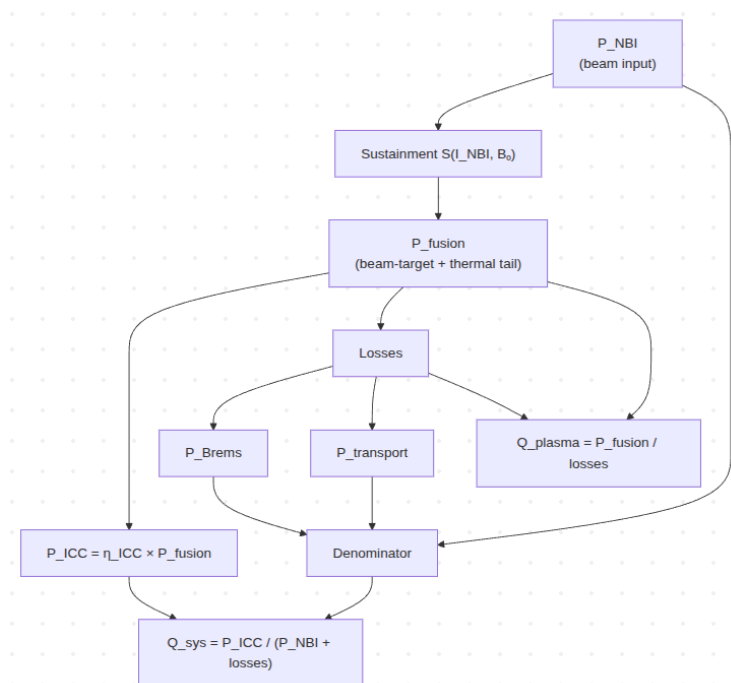
- **NBI Current (0–120 A, Default: 40 A):** Beam current (normalized). Sets beam energy, fast-ion fraction, input power P_{NBI} , and PIC injection rate.
- **Background B0 (0.1–5.0 T, Default: 1.5 T):** Peak $|B_z|$ at flat-top. Enters confinement time τ_E , bulk T_i , and core density.
- **ICC Coupling (0.50–0.95, Default: 0.85):** η_{ICC} : fraction of fusion power booked as electricity at the collector.

Beam sustainment (the physics the optimizer discovers)

TAE's Norm result is **NBI-only sustainment**: the reversed field is **created and held by beam-driven current**, not coils alone. The model uses a sustainment fraction $S(I_{NBI}, B_0)$:

- Below **~30 A**, $S \rightarrow 0$ — the FRC does not maintain reversal regardless of B_0 .
- **B_0 cannot substitute for beams.**
- Low NBI is legal but **self-penalizing** (τ_E collapses, transport losses rise).

There is **no hard floor** on beam current — the optimizer finds **~55 A** because physics demands it, not because of a constraint.



Power balance routing

Optimum from Optimize (current 0D model)

Pressing **Optimize** searches the 3D control space (coarse grid + local refinement, ~1 s, background thread). On the current codebase, the optimal operating values settle as follows:

- **NBI Current:** ~55 A
- **Background B0:** 5.0 T (slider maximum)
- **ICC Coupling:** 0.95 (slider maximum)
- **Q_{sys} (system gain):** ~0.066

$Q_{\text{sys}} = P_{\text{ICC}} / (P_{\text{NBI}} + P_{\text{Brems}} + P_{\text{transport}})$ — what must exceed 1 for net electricity.

The model is built to explore **proposed** p-11B operating physics on top of Norm-style beam sustainment. The optimizer's best point in this 0D parameterization still sits **below breakeven** — which is itself the lesson: even with beams, ICC, and aneutronic fuel, **Rider-class difficulty** does not disappear magically. Norm demonstrated **FRC sustainment**, not wall-plug $Q \geq 1$ on p-11B.

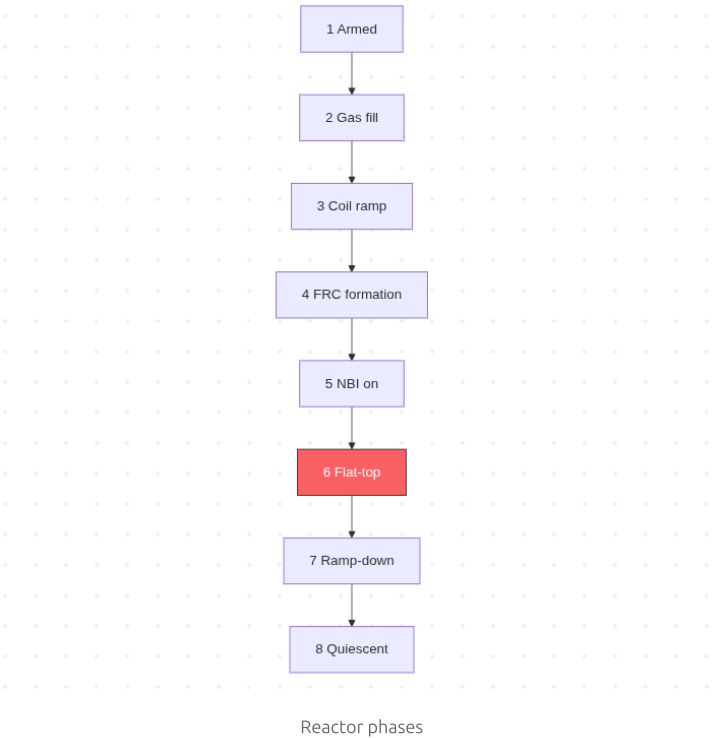
Output measurement graphs (right-hand diagnostic panel)

All three plots share a **time axis in microseconds (μs)** and update every physics step during the discharge.

- **1. Ion / Electron Temperature (top):** Monitors ion temperature T_i (red line) and electron temperature T_e (blue line) in keV. To bypass the Rider Limit, the goal is to keep ions hot (150–300 keV) while electrons remain relatively cold.
- **2. Core Power Balance (middle, log scale):** Measures fusion heating P_{fusion} (yellow line) calculated via $n_p n_B \langle \sigma v \rangle E_f$ where $E_f = 8.7$ MeV; relativistic Bremsstrahlung P_{Brems} (pink line) proportional to $Z_{\text{eff}}^2 n_e^2 / T_e$; and conduction/transport loss P_{cond} (green line) proportional to $n_e T_e / \tau E$.
- **3. Net Gain Q (bottom, log scale):** Measures plant gain Q_{sys} (solid white line) representing recovered ICC power against beam and transport losses; and physics gain Q_{plasma} (dashed yellow line) showing $P_{\text{fusion}} / (P_{\text{Brems}} + P_{\text{cond}})$ which excludes the beam injection power. A dashed grey line marks the $Q = 1$ breakeven point.

Reactor phases in the exported video (8 numbered callouts)

The MP4 walks through a scripted shot. Countdown labels are control-room shorthand — simulation time is compressed so you reach flat-top in seconds of real time, not minutes.



The sequence of operations during a simulation run proceeds as follows:

- **Step 1/8 - Armed:** Coils on standby, vacuum pumped, gas puffed (weak B_z).
- **Step 2/8 - Gas fill:** Boron and hydrogen inventory rises (green population builds) over approximately $0.8\ \mu\text{s}$.
- **Step 3/8 - Coil ramp:** The B_z colormap strengthens as the scaling factor reaches 1 over approximately $2.0\ \mu\text{s}$.
- **Step 4/8 - FRC formation:** Hot plasma macroparticles are seeded and the FRC topology appears over approximately $3.0\ \mu\text{s}$.
- **Step 5/8 - NBI on:** Fast red protons stream from the $-x$ side and beam heating begins over approximately $4.0\ \mu\text{s}$.
- **Step 6/8 - Flat-top:** Full discharge lasting approximately $25\ \mu\text{s}$. Includes fusion, yellow alphas streaming to the ICC, and active diagnostics.
- **Step 7/8 - Ramp-down:** Beams and magnetic field fall over approximately $4\ \mu\text{s}$.
- **Step 8/8 - Quiescent:** Plasma quiesces; shot complete.

What I take away

- 1. **p-11B is an engineering spectrum problem**, not just a triple-alpha stoichiometry problem — collection hardware cares about energy distribution.
- 2. **TAE's bet is structural**: beam-sustained FRC + separate boron fueling + ICC — our 2D slice makes that story visible.
- 3. **Simulation ≠ milestone**: Norm proved NBI sustainment; this GUI layers a **proposed** p-11B performance model on top — the optimizer still finds $Q_{sys} \ll 1$ in the current 0D parameterization.
- 4. **Open tools accelerate intuition** — colored macroparticles, synchronized narration, and diagnostic time histories beat equations alone for building mental models.

References

- 1. **TAE Technologies** — company program and public milestones: tae.com
- 2. **TAE Norm / NBI-only FRC sustainment (2025)**: *Nature Communications*, doi:10.1038/s41467-025-58849-5
- 3. **Rider, T. H.** — fundamental limits on fusion reactor systems (Rider limit): *Physics of Plasmas* (1995)
- 4. **Nevins, W. M. & Swain, M.** — p-11B reactivity and engineering constraints
- 5. **pb11_reactor_sim** source, tutorial, and physics notes: github.com/catskillsresearch/pb11_reactor_sim
- 6. **Catskills Research Company** — simulation disclaimer: models for theoretical research; not engineering specifications for physical deployment

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